The empirical recurrence for OEIS A296671: recovering a conjecture that is not written down

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Abstract

OEIS A296671 records "Empirical recurrence of order 67", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 125$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 67, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 67 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A296671 is "Number of nX5 0..1 arrays with each 1 adjacent to 0, 2 or 3 king-move neighboring 1s.". It has offset 1 and begins

$$13, 208, 1451, 16541, 213800, 2358963, 28341757, 346828018, \ \ldots$$

The entry states:

> Empirical recurrence of order 67 (see link above)

As of the "Last modified" line on the live entry (Dec 18 2017 11:03:56, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1056$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 125$ of the 1056, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 125$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 250$ exact terms of the model returns order 67 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{67} c_i\, a(n-i),$$

c_1	13	c_{18}	−11417442691	c_{35}	−169288880157217	c_{52}	27796352785490
c_2	−8	c_{19}	−37020124070	c_{36}	80703535983078	c_{53}	−20783047477069
c_3	288	c_{20}	−18871300450	c_{37}	181915418408314	c_{54}	−38801549460783
c_4	−3953	c_{21}	52934536615	c_{38}	138835932297780	c_{55}	4080914053186
c_5	184	c_{22}	184495748524	c_{39}	−41523126229421	c_{56}	−5349401287852
c_6	−14612	c_{23}	−6721080341	c_{40}	−375856329135729	c_{57}	12304256485016
c_7	235290	c_{24}	−534972517173	c_{41}	−114926336292678	c_{58}	−132858259386
c_8	434359	c_{25}	−1056482948604	c_{42}	16243658387966	c_{59}	−1248845984862
c_9	1205191	c_{26}	847326202152	c_{43}	266308838536622	c_{60}	−2369254914520
c_{10}	−5385771	c_{27}	5560579793890	c_{44}	374227610988309	c_{61}	384797834692
c_{11}	−24394434	c_{28}	3196408777580	c_{45}	−134486107122609	c_{62}	376383559592
c_{12}	−71096839	c_{29}	−6756961075016	c_{46}	−97904258136416	c_{63}	−47572389112
c_{13}	−31100226	c_{30}	−32646191945874	c_{47}	−207678430016995	c_{64}	−34921950736
c_{14}	397260209	c_{31}	20822645866474	c_{48}	−71532698402437	c_{65}	−54110592
c_{15}	1558885549	c_{32}	45888261528059	c_{49}	67552000244947	c_{66}	992155136
c_{16}	3038229185	c_{33}	25610380794722	c_{50}	157572059164426	c_{67}	74767360
c_{17}	−274618002	c_{34}	−53524315635722	c_{51}	−67867067799163		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 67, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $67 + 4$ terms removed returns order 67 as well, so $\deg q_{\min} = 67 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $67 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 67 that this sequence satisfies — and that family turns out to have exactly one member.

References

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